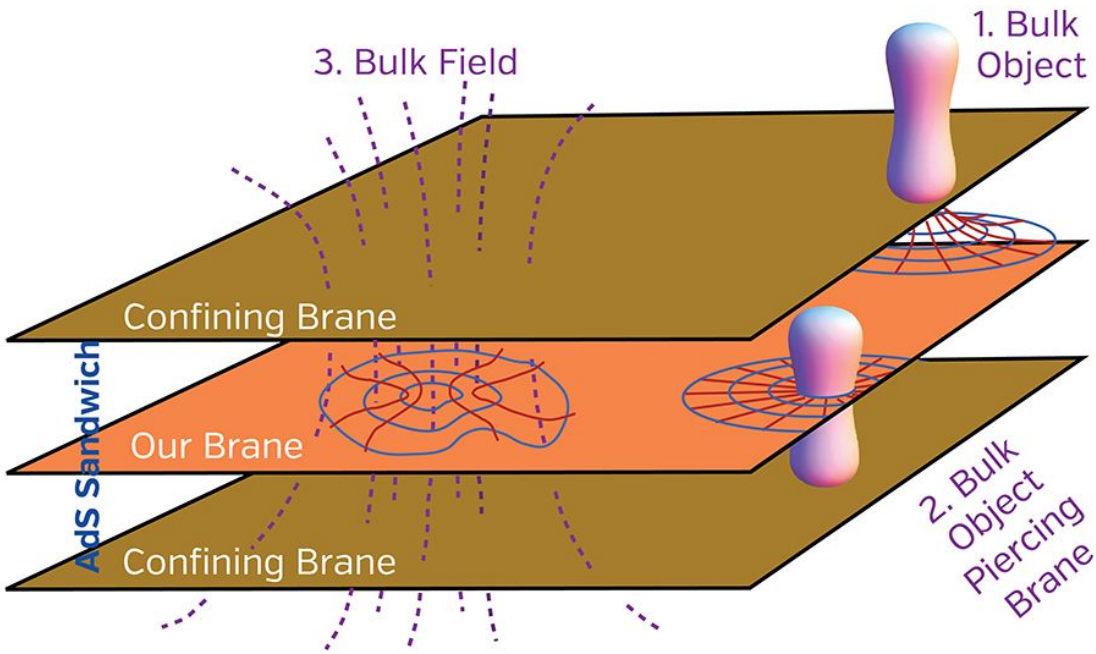


Fig. 25.3. Three ways that the bulk could produce the observed gravitational anomalies. The red and blue curves are tidal tendex lines produced by a bulk object or bulk field.

A bulk field is a collection of force lines that resides in the five-dimensional bulk. What kind of force lines, the Professor doesn't know, but he speculates; see below. Figure 25.3 shows a bulk field (dashed purple lines) passing through our brane. This bulk field generates tidal gravity in our brane (red and blue tendex lines). As the bulk field changes, its tidal gravity changes, resulting (the Professor thinks) in most of the observed anomalies.

But that isn't the only role of bulk fields, he suspects—in my extrapolation. They may also control the strength of the gravity produced by objects living in our brane, such as a rock or planet.

Bulk Fields Control the Strength of Gravity

The gravity of each little bit of matter in our brane is governed, to high accuracy, by Newton's inverse square law (Chapters 2 and 23): its